

Review

A review of the application of coral reef condition indicators in conservation planning

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ABSTRACT

Monitoring data have shown declines in the condition of coral reefs. However, it is unclear how such monitoring data are used to inform spatial conservation planning decisions to slow and prevent further declines. At least six of the Kunming–Montreal Global Biodiversity Framework Targets (1, 2, 3, 8, 14 and 21) would be improved if coral condition monitoring was better connected to spatial conservation planning. Here, we review published literature for coral condition monitoring in coral reef conservation planning to identify why, how, and which indicators are used to determine coral reef condition. Across 219 monitoring studies, we found 159 different indicators have been used to monitor coral reef condition. While 48% of the studies recommended the use of monitoring data to improve conservation outcomes, only 10% used condition data to inform a spatial conservation plan, and 14% used condition data to measure a plan's impact. Although monitoring data are useful for improving conservation outcomes, they are rarely applied in conservation planning. By synthesising the existing data and protocols, we provide recommendations of how to address and change the current disconnect between coral reef monitoring and informing spatial actions focusing on data democratisation to see increased data in decision making.

1. Introduction

Coral reefs are among Earth's most diverse and productive ecosystems, providing habitat for approximately 25 % of all marine life (Green and Bellwood, 2009; Hughes et al., 2017) and ecosystem services for hundreds of millions of people (Eddy et al., 2021; Pellowe et al., 2023; Santavy et al., 2021). However, coral reefs are threatened by numerous local (e.g. overfishing, land-based run-off) and global (e.g. ocean warming and acidification) impacts (Brown et al., 2013; Goldberg and Wilkinson, 2004; Cornwell et al., 2021). These cumulative and escalating impacts have resulted in a 50 % reduction in global coral cover between 1957 and 2007 (Eddy et al., 2021) and a 14 % decline between 2009 and 2018 (Souter et al., 2020). Statistical and process models estimate that 90 % of coral reefs will be lost by 2050 unless the rate of anthropogenic climate change is mitigated and adequate investment is made in local management actions (Hughes et al., 2018; McClanahan et al., 2009; Wilkinson, 2006).

The Kunming–Montreal Global Biodiversity Framework (GBF) of the

UN Convention on Biological Diversity (CBD, 2022) seeks to halt and reverse biodiversity loss, specifically addressing the coral reef crisis through global targets for protection (Target 3), restoration (Target 2), enhanced resilience (Target 8), and improved data utilisation to inform management (Targets 1, 14, and 21). Marine conservation planning is crucial for achieving GBF Target 1, which calls for systematic conservation planning to identify priority areas and actions to enhance coral reef resilience (Beyer et al., 2018; Douvère, 2008; Gudka et al., 2024; Lopazanski et al., 2023; McClanahan et al., 2024; McLeod et al., 2021). Spatial data on the extent and condition of coral reefs is core to systematic conservation planning. Furthermore, effective monitoring is necessary to ensure that actions taken towards the GBF targets result in the expected outcomes.

The adaptive management paradigm constitutes an integral part of systematic conservation planning for coral reef ecosystems. It integrates monitoring data into a framework that reconciles management outcomes with ever-changing socioeconomic and conservation objectives (Day, 2008; Douvère and Ehler, 2011; Poiani et al., 1998; Reynolds

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et al., 2011). The Great Barrier Reef Marine Park (GBRMP) is a successful example of effective adaptive management in action. Every decade the GBRMP plan is evaluated to examine if targets, objectives, and outcomes are met. This is measured with monitoring data, which is used to inform future management and potential rezoning (Great Barrier Reef Marine Park Authority and Queensland Government, 2015). One example of this is the management of crown-of-thorns starfish outbreaks. Using a targeted prioritisation approach of monitoring and action, crown-of-thorns starfish culling has delivered widespread benefits for the entire reef system (GBRMP Reef Authority, 2025).

Monitoring, particularly long-term monitoring, which involves repeated measurements collected for at least ten years (Mills et al., 2023), is fundamental to adaptive management, systematic conservation planning, and achieving the goals of the GBF by ensuring the ‘right’ places are protected, such as areas with high biodiversity or ecological importance. Monitoring is embedded in every stage of adaptive management, enabling the quantification of impacts and responses of key ecological processes (Lindenmayer & Likens, 2010) (Fig. 1). Without long-term monitoring, adaptive management is difficult and lessons from past management cannot be evaluated and used for improvements (Angermeier and Karr, 2018; Field et al., 2007; Lindenmayer et al., 2013; Nichols and Williams, 2006).

Biodiversity monitoring is essential for adaptive management and

underpins the success of the GBF. However, the path to achieve Target 21 “ensure the best available data, information and knowledge are accessible to decision makers, practitioners and the public to guide effective and equitable governance, integrated and participatory management of biodiversity, and to strengthen communication, awareness-raising, education, monitoring, research and knowledge management” (GBF, 2022) is less clear.

As there is no internationally agreed-upon set of guidelines for what or how to monitor coral reefs, it is unclear how to apply monitoring data to guide effective management decisions. This is partly due to a lack of understanding of the important components of coral reef ecosystems that show changes in the quality of the ecosystem over time (Borja et al., 2012). As a result, time and money, both limited resources, are invested in monitoring efforts that do not align with spatial planning objectives. When conservation plans ignore, or rely on poor-quality, opportunistic, or anecdotal data (McDonald-Madden et al., 2010), they risk misplacing management actions (spatial planning), thereby undermining their effectiveness (Day, 2008; Dunstan et al., 2016; Leslie, 2005; Lundquist and Granek, 2005). This disconnect between data collection and application is a major obstacle to achieving GBF Target 21, which calls for accessible, high-quality data to support equitable and effective biodiversity governance.

Here, we aim to synthesise what indicators are used to monitor and measure coral reef ecosystem health, and how these have been used in

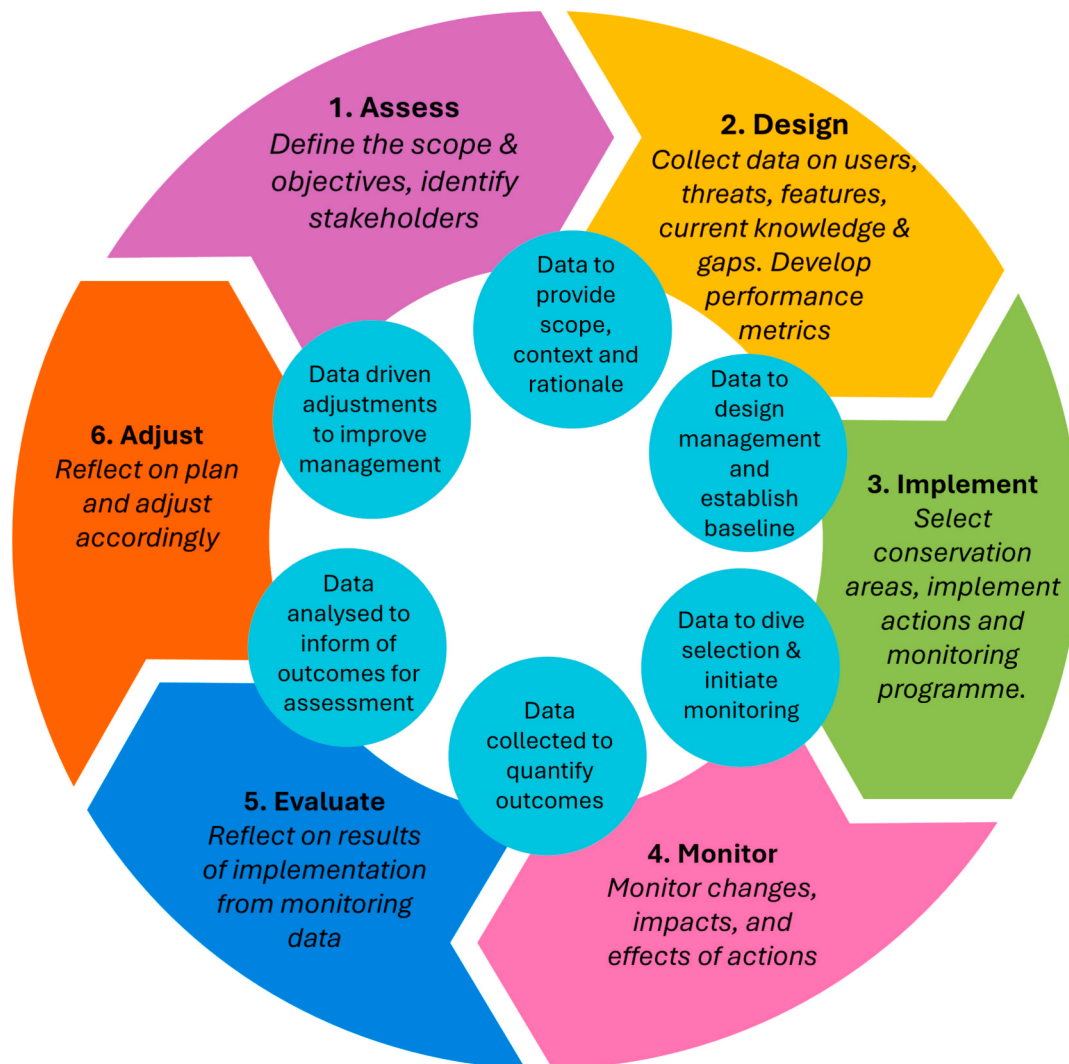


Fig. 1. The adaptive management cycle specific for conservation planning. The main opportunities for monitoring data to inform conservation planning is shown in the smaller inner (aqua coloured) circles (adapted from (Williams and Brown, 2018)).

conservation planning. To better understand the indicators that define coral reef ecosystem conditions and how effectively biodiversity data are used in conservation planning for coral reefs (GBF, 2022), we reviewed studies that used indicators for coral reef condition. Our objectives were to develop: 1) an understanding of the indicators commonly used in coral reef monitoring; and 2) pathways for applying condition data in spatial planning for coral reef conservation.

2. Methods

To assess the approaches used to measure coral reef condition, we conducted a review of publications pertaining to coral reef condition using the ISI Web of Science (as of June 2024). Our focus on the term condition was grounded in its relevance to an ecosystem's capacity to

maintain fundamental ecological functions and sustainably deliver ecosystem services and resources—making it highly relevant to conservation planning decisions and outcomes (Borja et al., 2012; Kandziora et al., 2013; Müller, 2005; Rapport, 1989). We used the following search topics: 'coral reef condition indicators', 'coral reef health indicators', 'measuring coral reef condition', 'measuring coral reef health', 'coral reef monitoring data', 'monitoring coral reef condition', 'monitoring coral reef indicators', 'data coral reef condition', and 'data coral reef health'. To understand how coral reef condition indicators are used in conservation planning, we conducted an additional search using the same keywords for condition above, paired with 'marine spatial planning', 'decision science', 'MPA design', 'reserve design', 'spatial conservation', 'systematic conservation planning', 'prioritisation', 'protected area design', and 'conservation planning'.

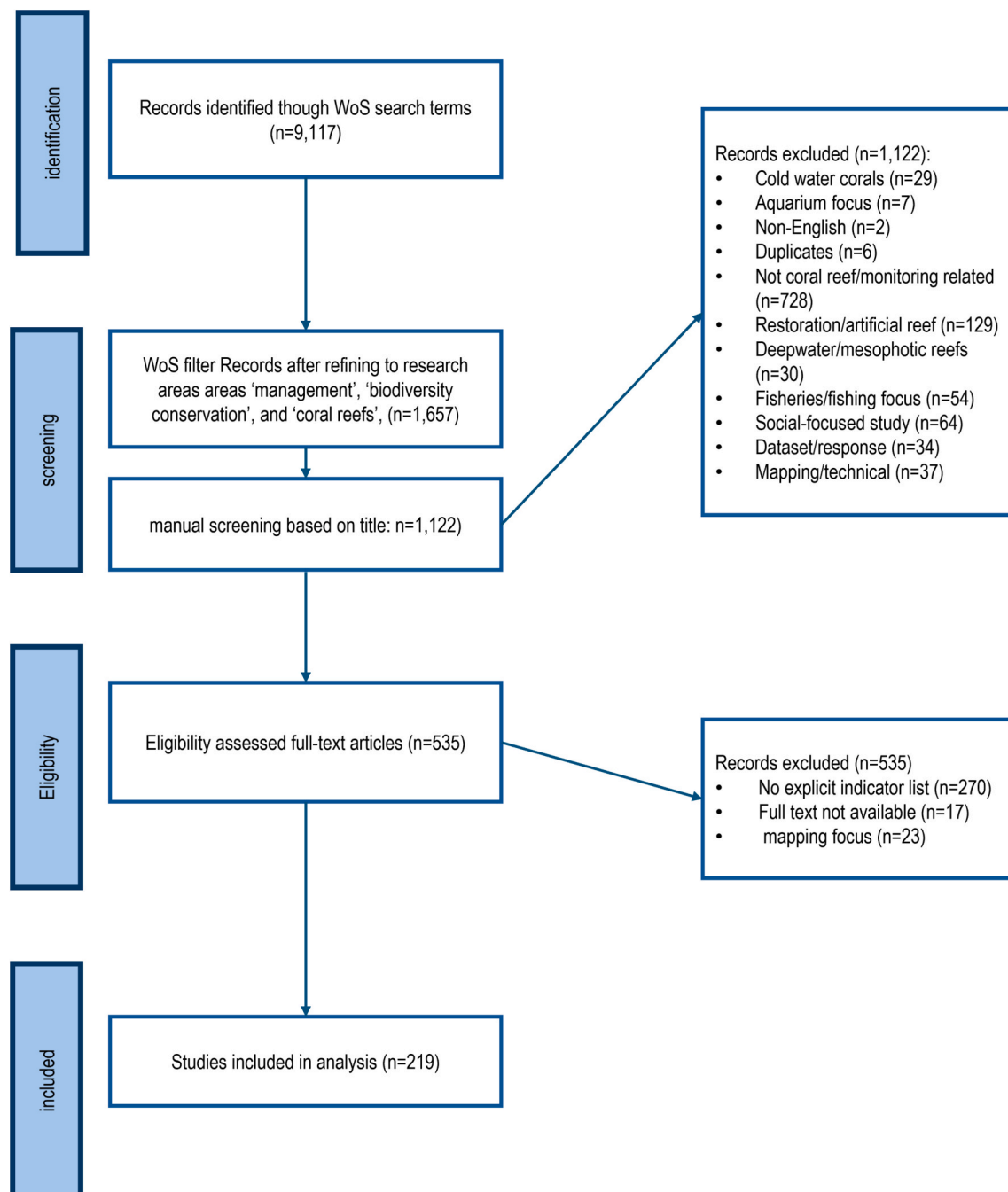


Fig. 2. Our search and refinement process for the literature review undertaken.

These searches initially returned 9,117 studies. Using the Web of Science “refine” function, we refined these to the research areas ‘management’, ‘biodiversity conservation’, and ‘coral reefs’, returning 1,657 studies. We recognise that there are many grey literature publications that we did not assess (i.e., reports and papers that have not been peer reviewed, such as nongovernment organisation reports and citizen science programs). We only used English language articles because of our lack of non-English language skills. We recognise this deficiency noting that 35 % of conservation studies are in languages other than English; this is especially problematic for Asia, Africa and Latin America (Amano et al., 2016, 2023; Bayraktarov et al., 2020).

To meet our inclusion criteria, studies had to explicitly mention coral reef condition indicators in their methods or results sections. Indicators were defined as measurements of environmentally relevant attributes used to depict or evaluate environmental impacts, conditions, or changes (Heink and Kowarik, 2010). As we focused on identifying different approaches to measuring coral condition, we only included indicators from specific monitoring programs once (i.e. common programs, such as ReefCheck, which is a global citizen science monitoring project and has been referenced in multiple studies). After removing duplicates, cold-water coral studies, and non-relevant articles, we reviewed 219 studies (Fig. 2).

From each article, we classified and grouped indicators with respect to broad purpose and specific reasons the study took place. The *purpose* for the monitoring was categorised into four categories: 1) no mention of the intended use of monitoring data – i.e. no explicit purpose for the application of the data, 2) mention of the benefit of monitoring for conservation/management use, 3) monitoring the results of a marine conservation plan, and 4) direct application to a spatial conservation plan design. We further classified the *reason* for monitoring into six categories, on: 1) to choose the most appropriate management actions; 2) to learn about the ecosystem to make informed choices in the future; 3) to raise awareness for public engagement; 4) to influence policy; 5) to discover without a set plan for future data use; and 6) to test or validate new tools, techniques, or methodologies (Lindenmayer and Gibbons, 2012).

Indicators were then grouped into three broad categories based on the attribute used—physical, socio-economic, and ecological. We also recorded the duration of each study, as well as the frequency and of monitoring. To assess the alignment of monitoring efforts with the goals of GBF Target 21, we compared studies that cited conservation or management benefits as their primary motivation with those that directly applied monitoring data to conservation plan design.

3. Results

A total of 219 studies published between 1976 and 2024 identified 159 distinct indicators used to measure coral reef condition. Of these studies, 17 % (n = 37) were long-term (indicators measured at various intervals for > 10 years). With respect to the monitoring purposes, 10 % (n = 22) of the reviewed studies applied condition indicators directly to marine conservation plan design or included spatially explicit management recommendations (purpose 4). Of the 219 studies, 14 % (n = 30) studies used indicators to monitor the outcomes of a conservation plan (purpose 3), while 48 % (n = 106) mentioned the benefit of monitoring for conservation or management use (purpose 2). There were 28 % (n = 61) studies with no mention of how the monitoring data would be used (purpose 1).

This distribution was similarly reflected in the reasons provided for the outcome from monitoring. The most common reason—cited in 44 % of studies (n = 96) was to learn about the ecosystem to inform future decisions (reason 2). As expected, studies that did not mention conservation applications were typically designed to explore or test new monitoring methods or tools (reasons 5 and 6) (Fig. 2).

3.1. Indicators

To understand coral reef conditions that are frequently monitored, we divided the indicators into three broad classes: physical, socio-economic, and ecological. Ecological indicators were the most frequently used, with most indicators measuring the benthic cover types (e.g. percent algae cover) and the relative condition of the biotic benthos (e.g. coral cover diversity and percent living). Ecological indicators can be divided into two general groupings to describe the condition: 1) species/habitat cover, abundance, and diversity, which can indicate positive, descriptive (i.e. presence, but positive or negative changes are less clear), or declining conditions and 2) threats – negative, direct, ecological impacts, pressures, or responses such as invasive species or coral bleaching cover.

Indicators that align with positive metrics of coral reef health, with higher cover and counts indicating a healthier reef state included live coral cover, which was the most frequently cited metric, used in 81 % of papers. Coral cover was measured in multiple ways, including total live coral cover (n = 127), soft coral cover (n = 38), coral cover per family, species/genus, and growth form/morphology (branching, boulder, etc.) (n = 89), coral richness (number of different corals) (n = 35), and coral diversity (n = 58). Other positive coral metrics included coral recruitment and juvenile density (n = 31), and coral growth rates (n = 9).

Metrics that indicated poor ecosystem health for coral health and demographic indicators, which are often used in conjunction and comparison with positive indicators included coral mortality or dead-to-live coral ratios (n = 76), coral bleaching (n = 65), coral disease (n = 35), coral damage (n = 12), predation or bite rates on coral (n = 6), and coral rubble (n = 35), (Fig. 3a).

Indicators used in monitoring that are descriptive of the ecosystem with no clear condition indicator, or where the positive/negative metric is outcome dependant, were focused on the benthic habitat cover. Algal cover was the most frequently used indicator in this sub-category, used in 105 studies, which, depending on the cover amounts, algae type, can be positive or negative. Other benthic habitat classifications included sand (n = 50), rock (n = 26), seagrass (n = 24), sponge (n = 36) cover metrics, consolidated hard substrate/pavement (n = 20), and 15 mentions of benthic habitat as an indicator without specific cover types included in the study.

For marine fauna, measures of fish diversity and abundance, where higher numbers are a positive indicator for coral reef condition were the most common – fish density and biomass were used in 18 studies; fish functional and/or taxonomic groups were mentioned 25 times, and species abundance mentioned 29 times. Specific focus was given to chaetodonts – (butterfly fish) a fish family that are corallivores (relying on live coral for food) (n = 7), herbivorous fish (n = 20), and other species-specific mentions. Focus on specific fish species is particularly relevant in conjunction with fishing pressure, which was a frequent socio-economic indicator. Additional descriptive, or context specific fauna indicators included anemones (n = 7), seashells (n = 8), and giant clams (n = 6). Negative, ecological indicators (threats) focused on species which are known to reduce coral reef condition when numbers exceed sustainable limits, including Crown-of-Thorns starfish (n = 20) and drupella (n = 3) – both of which are corallivores and are known to have ‘outbreaks’ in population numbers, urchin abundance (n = 16), and invasive species (n = 7). (Fig. 3b).

There were fewer physical/abiotic indicators than biological/biotic. Physical indicators focused on direct (i.e. sedimentation) and indirect (i.e. sea surface temperature) threats that alter the ecosystem health. The most common was sea surface temperature (n = 57). Water quality indicators were a major focus and included chlorophyll-a (n = 26), water transparency and turbidity (n = 36), salinity (n = 25), pH (n = 14), sedimentation/silt rates (n = 40), and runoff (n = 4). Weather and oceanographic indicators included cyclones/hurricanes (n = 19), disturbance events (n = 11), temperature and variability (n = 18), and wave impact/intensity (n = 21) (Fig. 3c). Physical indicators of coral

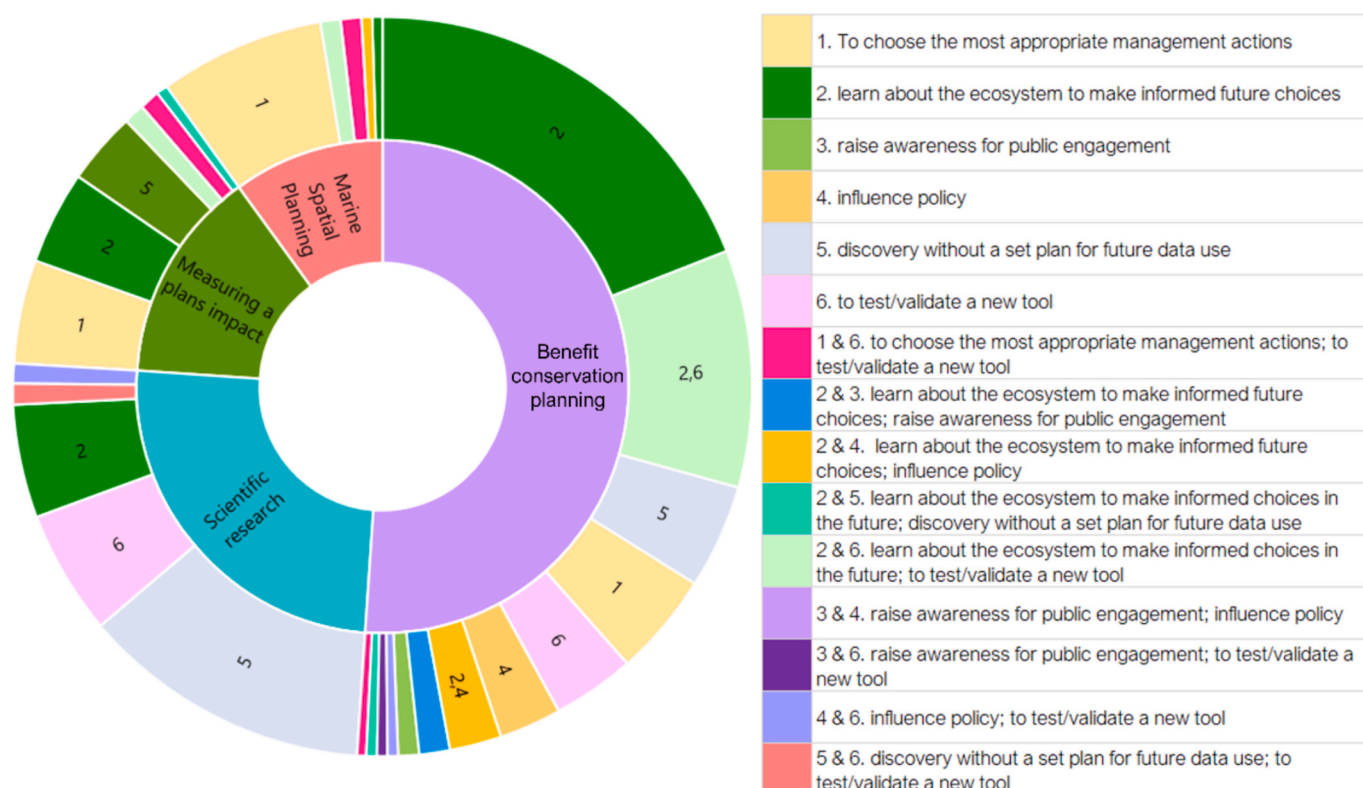


Fig. 3. The broad purpose for monitoring based on what is stated in the article (inner circles, colour-coded and proportional to the number of papers per category) and the specific reason the monitoring took place, shown as the second (outer) layer circle proportional to the inside circle (purpose) and described in the legend.

reefs, which could be a proxy for meeting representation goals within a planning region, included reef depth ($n = 40$), geomorphic zone ($n = 35$), rugosity ($n = 32$), as well as reef slope, zone, and exposure ($n = 16$).

In the socioeconomic indicator group, the most common metrics were used to monitor direct uses and pressures on the ecosystem, with the most frequent indicator being the type, level, and age of protected area management (i.e. no take) ($n = 43$), followed by fishing pressure/impact and duration ($n = 31$). Both are related to fishing levels, as protected areas generally disallow or manage fishing. Other fishing-related indicators included destructive fishing practices ($n = 11$), commercial fishing catch ($n = 8$), and fisher population/density ($n = 4$), which are attributable to region-specific fishing pressures. Some threat indicators focused on adjacent uses and considered users of the space. These included human population, urban density ($n = 24$), and tourism levels ($n = 18$). Monitored land-based pressures included pollution ($n = 11$), land use/agriculture/farming ($n = 8$), and urban area/development ($n = 12$) (Fig. 3d).

3.2. Applications of monitoring

To assess how monitoring data are applied in biodiversity conservation in line with GBF Targets 21, we compared the indicators used in studies that aimed to benefit conservation and management (purpose 2) with those publications that directly applied indicators in a conservation plan (purpose 4). We found that over half (55 %) of the indicators that aimed to benefit conservation planning were used in studies that applied monitoring data to conservation planning. However, many of the studies that reported the usefulness of indicators for future conservation planning did not specify whether a conservation plan was being implemented in the study area. Notably, we did not find any studies that used data collected in a subsequent conservation plan. There were 27 ecological, 23 physical, and 18 socioeconomic indicators that were used to measure the impact of conservation plans, or to demonstrate

conservation benefits, but not applied in spatial planning (Fig. 4).

When comparing the frequency and purpose of coral reef monitoring studies and the introduction of new indicators over time, we found a divergence in trends. The number of studies monitoring coral reefs has steadily increased, particularly since 2015. However, the number of new indicators introduced each year has plateaued or declined since around 2010. With respect to purpose, early publications (1970's–2000) were primarily focused on scientific discovery. From 2000 onwards, there was a noticeable shift towards studies aimed at conservation outcomes, including conservation planning and the evaluation of conservation plan impacts. This trend suggests that while interest in coral reef monitoring continues to grow, the field is increasingly relying on a core set of established indicators rather than developing new ones (Fig. 5).

4. Discussion

Coral reefs are complex and variable in time and space (Hatcher, 1997) and it is difficult defining what constitutes a 'healthy' reef given current knowledge gaps. This makes it difficult to measure coral reef resilience, recovery, and responses to change, and at the same time meaningfully apply associated metrics in conservation planning efforts. Despite the number of indicators found in our review of 219 coral reef monitoring studies, we found a significant disconnect between data collection and the application of that data in conservation planning. Nearly half (48 %) of the studies were designed with conservation benefit purpose, yet only 10 % of the reviewed studies applied monitoring data to plans, and just 14 % using monitoring to measure plan impact.

Our review highlights the rare use of monitoring data to inform critical conservation decisions, despite the volume of published monitoring studies. This implementation gap persists despite the GBF placing biodiversity data at the centre of action targets, threat identification, priority setting, benchmarking, and progress reporting (GBF, 2022). In

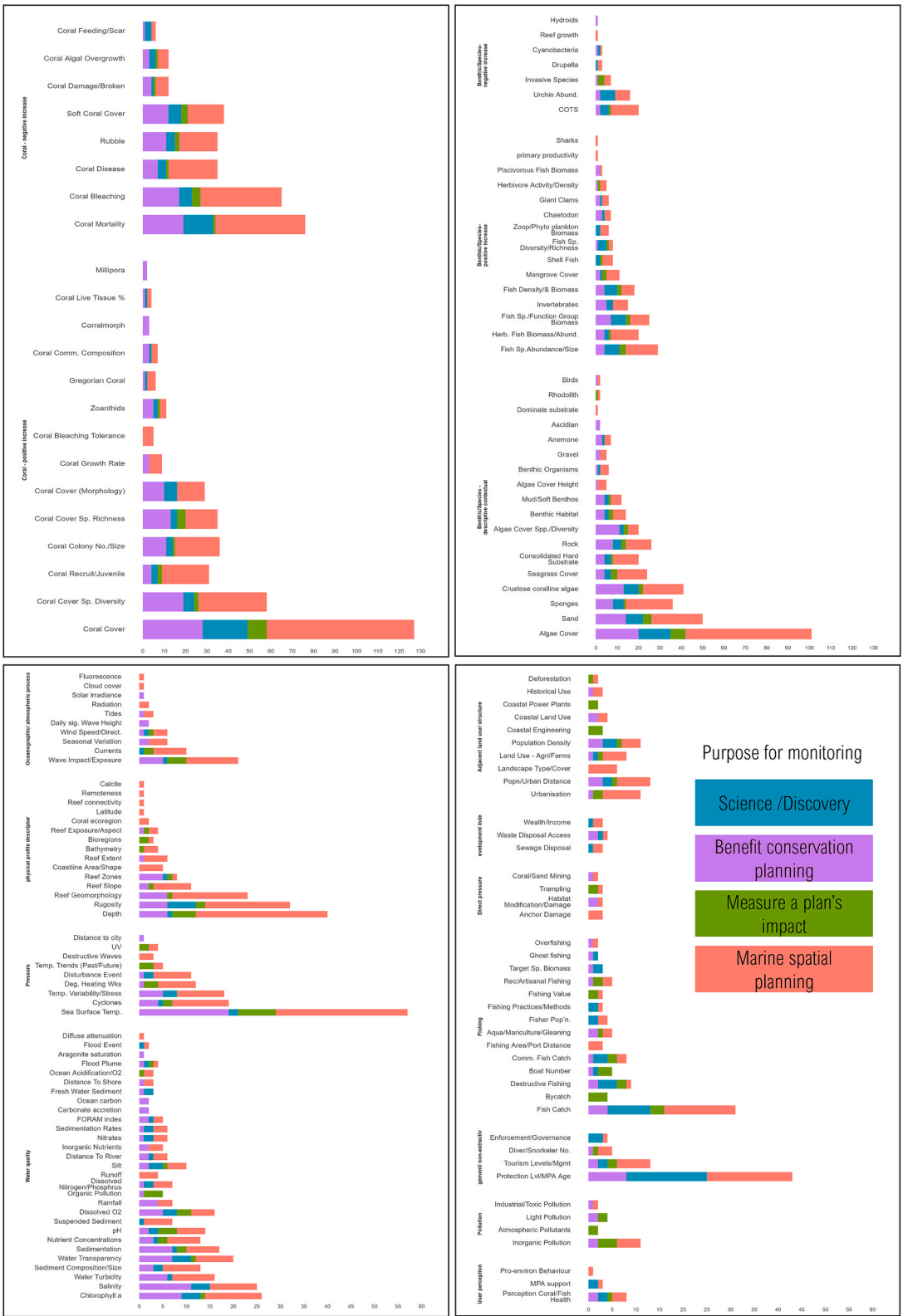


Fig. 4. The number of mentions and types of indicators reviewed and grouped into the four paper categories: a, coral cover, broadly classed in positive and negative based on measured increases to the specific indicator for reflecting coral reef condition. b, other ecological and benthic indicators, grouped in positive or native measured increases, and descriptive or context specific. c, physical, and d, socioeconomic, both grouped by overarching information the indicators reflect. Indicators are presented by the purpose stated for monitoring as colours in the bar graph (scientific research in blue, measuring a plan's impact in green, beneficial for conservation planning in purple, and marine spatial plans in pink. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

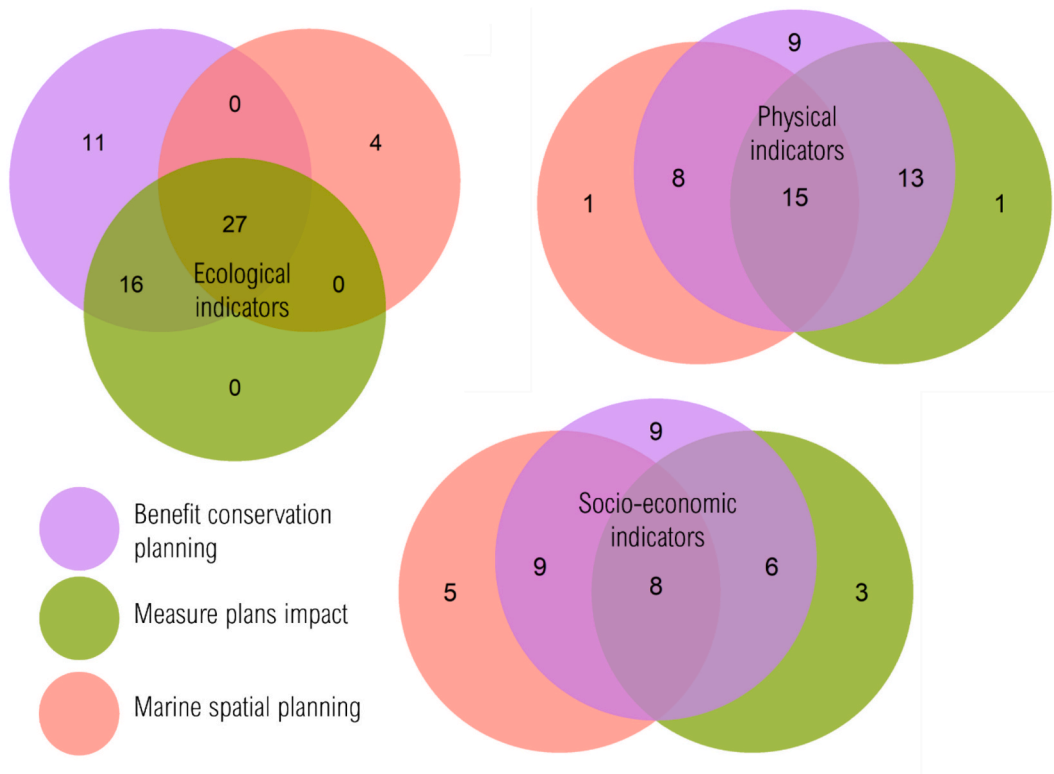


Fig. 5. Venn diagrams showing overlapping and unique indicators based on the stated purpose of the study: used for marine spatial planning (purpose 4, pink), to provide other benefits for conservation planning (purpose 2, purple), and to measure the impact of management (purpose 3, green). Numbers represent the number of indicators that are used in the 3 purposes for monitoring, and their respective uses across multiple purposes, with the centre number representing indicators used in all purposes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

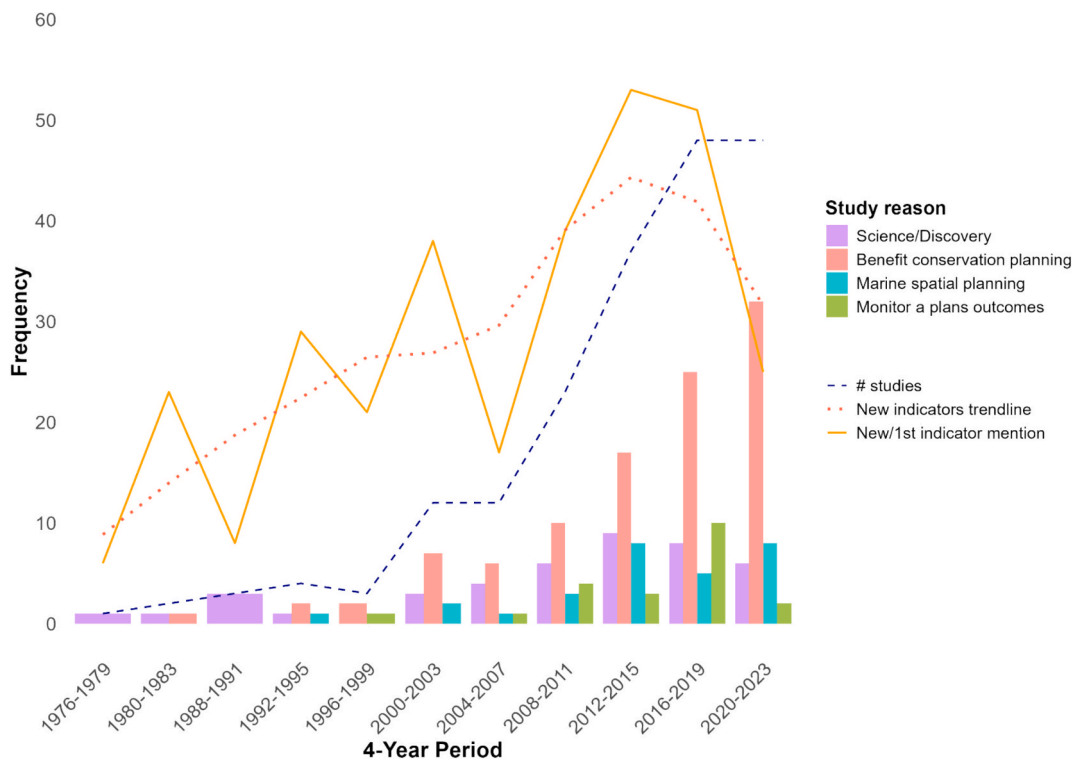


Fig. 6. Frequency plot showing the number of papers published per reason category per year period (bars, colour-coded with reason for the study); the total number of papers per 4 years (blue dashed line), the number of new indicators (orange line), and new indicator trend line (red dotted line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

practice, monitoring indicators are rarely integrated into conservation planning design despite data being central to successful systematic conservation planning.

Conservation planning aims to identify areas that efficiently and effectively protect biodiversity and/or ecosystem functions while minimizing negative impacts to other users (Margules and Pressey, 2000). Its success depends on the use of relevant data to guide decision-making (Binley et al., 2024; Gerber and Iacona, 2024). However, the most effective indicators for monitoring coral reefs—and how they can inform conservation decisions—remain unclear. This is partly due to context-specific needs, but our review also highlights this is because existing indicators are rarely applied in practice.

Although there are knowledge gaps in how to measure coral reef ecosystem functioning, this alone does not explain the disconnect between data collection and application in conservation planning. Our analysis reveals that large volumes of data are being collected that has clear benefits for conservation planning—such as fish diversity and abundance, fishing levels, and benthic habitat classifications. These metrics could play a key role in conservation design and ongoing monitoring for adaptive management, yet the implementation gap remains. (Fig. 6).

While no universal criteria exist or became evident from our review for a ‘good’ indicator list, best practices can be applied to ensure monitoring data is useful for conservation.

For monitoring data to be useful for conservation it must be measurable across time and space and be able to address key planning questions. These include identifying the location, distribution, and amounts of the features targeted by the plan’s objectives; determining what and where the main conflicting uses and users of the planning region occur and at what extent; assessing the distribution, magnitude, and intensity of threats that can be mitigated and clarifying which threats can be addressed where; and, for threats that cannot be mitigated, understanding their impacts and considering whether sites with very high unmitigable threats should be avoided in a plan.

Many of the indicators we reviewed are useful in providing answers to these conservation planning questions. But their utility relies on their dissemination and application. In essence, a ‘good’ indicator is salient in both the environmental characteristic it is measuring and its usefulness and accessibility for decision making. We argue to promote increased uptake of these data into conservation planning, data collection, maintenance, and management should consider the following three attributes to be better used in conservation outcomes:

4.1. Data consensus

Indicators should accurately describe and measure ecosystem components, but they must also be comparable across studies to expand the scale and applicability of monitoring efforts. Our review found that while many studies used similar metrics, they employed different sampling methods and reporting formats. For example, hard coral cover was measured as a ratio of live to dead coral or categorised by morphological classes, depending on the study. This inconsistency hampers cross-study analysis and prevents planners from consolidating data, as metrics cannot be reliably aligned.

The lack of consensus on which indicators to prioritise complicates conservation decision-making (Carpenter et al., 2008; Likens and Lindenmayer, 2018; Robinson et al., 2018). Even the most frequently used indicator, coral cover, is contested in its usefulness and methods of monitoring. This is because high coral cover may be from a single, or a few dominant species that grow rapidly after disturbance (such as mass bleaching), which may not reflect a healthy, biodiverse ecosystem (Brito-Millán et al., 2019; Hochberg and Gierach, 2021; Richards, 2013). Establishing a minimal global indicator set and standardised measurement protocol, supplemented by region-specific metrics, would improve data comparability and support consistent conservation planning worldwide.

4.2. Ensure data can be accessed and used by all

The need to democratise data sharing for conservation decisions is clear. Even when datasets exist, they seldom reach decision makers in a usable and consistent form. Inconsistencies in data organisation, lack of standardised metrics, and incompatible methodologies make it difficult to compare and apply data across locations and scales (Reynolds et al., 2011). The mismatch between the number of papers aimed to benefit conservation planning, and those applying monitoring to conservation actions highlights the need to improve information transfer and sharing in formats that are useful for practitioners. Our review found that large volumes of data are generated, however these are often difficult to analyse, replicate, or apply meaningfully to conservation planning, and additionally are difficult to analyse in the short-term and too difficult and expensive to replicate in the long term (Álvarez-Romero et al., 2018). This is often a result of a lack of access to data, lack of methods, or complex analysis.

The lack of data accessibility has led to marine spatial planning studies relying on global datasets that focus on pressures and threats rather than key ecological and physical attributes and values, meaning decisions for conservation are made on threats, not values. For example, two marine spatial planning studies used the human impact footprint data, which was not used in any other paper categories, while coral disease metrics were used in 22 studies aiming to benefit conservation, but only one study used this for a spatial plan.

The barriers for data democratisation arise from multiple aspects; preparing data is time consuming, costly and underfunded, and often attached to risks. However, recognising the urgency to implement coral reef conservation actions, it is arguably more beneficial to share what data exists then to restrict access, with the goal of Findable, accessible, interoperable, and reusable (FAIR) data (Trice et al., 2021). An example of data and method access is the Reef Life Survey data and health indicators, which provides ecological data which could be used in conservation planning. The data is available to download freely, and the methods and rationale of analysis are freely available, meaning the data, rationale, and methods are available to apply.

Achieving GBF Targets 14 and 21 requires a stronger emphasis on data management protocols and standards that are suitable and necessary for management, not just on expanding streams of more and large datasets. This will improve transparency, reproducibility, and the potential for broader application, fundamental principles of systematic conservation planning. This is linked to the abovementioned data transfer, access, and consolidation to ensure consistency and interoperability across different methods and programs that measure the same indicators.

Additionally, making practitioner and user case studies, examples, and evaluations of marine spatial planning exercises more common in scientific publications could provide better clarity around how monitoring data is used, helping to tailor the needs for transfer, access, and types of information. Increasing case study information on data use would enable better approaches to adaptive management, inform scientists on what and how data is used to inform decisions, and potentially reduce the data mismatches and redundancy (McIntosh et al., 2018), leading to dual science and practitioner benefits.

4.3. Ensure data is able to be scaled up for a planning region

Temporal and spatial mismatches in monitoring data present a significant barrier for effective use of data in conservation planning. Long-term monitoring is essential to detect ecological trends and temporal changes in biodiversity (García-Barón et al., 2021), yet only 17 % of reviewed studies were long-term (>10 years). This shortfall is often due to a lack of explicit goals, weak survey design, and limited statistical power. While we reviewed many robust high-quality examples of coral reef monitoring, there is inconsistency in what types of information, time scales, and the appropriate spatial size that are useful for

conservation planning.

The issue of scaling up and meaningfully applying point and other raw data for large-scale planning is pervasive. Scaling up point data requires high-level expertise in specific tools – for example species distribution modelling. This is especially challenging in developing and non-English-speaking countries, where coral reefs are more abundant but financial support and access to technical expertise are limited (Johri et al., 2021). Research is disproportionately concentrated in the global north, negatively correlated with coral species richness, and unrelated to threats to coral reefs globally. This indicates a serious mismatch between conservation needs and the knowledge required for effective monitoring and management (Fisher et al., 2011). Sharing of knowledge in applying data is needed to improve the usefulness of data.

5. Conclusion

While there is no solution to solving the mismatch we found between the number of monitoring studies and the number of conservation plans that use these data, best practices exist to improve data use in conservation plans. Alongside data structure, availability, and scalability, there needs to be long term investment in funding, training, and integration. Monitoring is a key priority for evaluating ecosystem conditions, but it is costly and prohibitive, especially in marine environments. Despite an increase in the number of species and habitats threatened or at risk, already limited funding and resources for conservation and monitoring are declining (Waldron et al., 2013). Fragmented and limited funding long-term monitoring programs has led to opportunistic and often reactive monitoring studies – often triggered by disturbance events like mass bleaching. These result in data that are patchy in space and time. Better funding for data collection, management, and sharing is needed to improve access and use for conservation.

We note that the bias towards data collection in more developed countries is a known limitation in coral reef research (Roch et al., 2025) and is also a limitation of the analysis in our review methods. We recognise our study only used peer-reviewed English-language articles, meaning we excluded a range of additional resources available, with 35 % of conservation studies in languages other than English (Amano et al., 2016, 2023; Bayraktarov et al., 2020), often developing nations. By focusing our study on English language peer-reviewed literature, we have excluded a wide range of resources available for coral reef monitoring. Additionally, our exclusion of grey literature is a limitation. There are multiple publications (i.e the International Coral Reef Initiative (ICRI), Atlantic and Gulf Rapid Reef Assessment (AGGRA)) that provide coral reef monitoring frameworks and indicators. Because of difficulties in searching for these, we excluded them from our analysis but acknowledge the value these programs provide with broader perspectives, local initiatives, and in-action projects. A small comparison of some of these programmes did reveal a large cross over with the indicators listed.

Owing to the increasing pressure on coral reefs, the challenges in identifying, maintaining, and protecting ecosystem functions crucial for coral reef health and the millions that rely on them are greater than ever (Bellwood et al., 2019). To better integrate reef monitoring into conservation planning, monitoring programmes must consider not only the importance and salience of data collection, but how data are transferred, scaled, and applied to real-world decisions. Addressing the best practice considerations for improving data usefulness will improve progress toward at least six GBF Targets (1, 2, 3, 8, 14, and 21). Better application of monitoring data in conservation planning will allow for more dynamic, transparent, and comprehensive approaches to conservation planning, allowing decisions to be data-driven, values based, and adaptable to the dynamic threats coral reefs face.

CRedit authorship contribution statement

Madeline Davey: Writing – original draft, Visualization,

Methodology, Investigation, Data curation, Conceptualization. **Carissa Klein:** Writing – review & editing, Supervision. **Chris Roelfsema:** Writing – review & editing, Supervision. **Caitie Kuempel:** Writing – review & editing, Project administration. **Hugh Possingham:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecolind.2025.114222>.

Data availability

No data was used for the research described in the article.

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